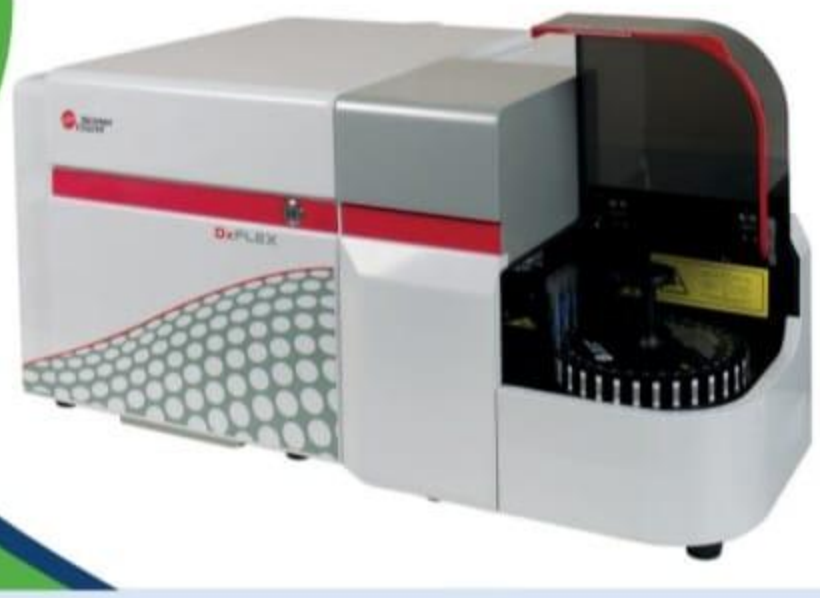
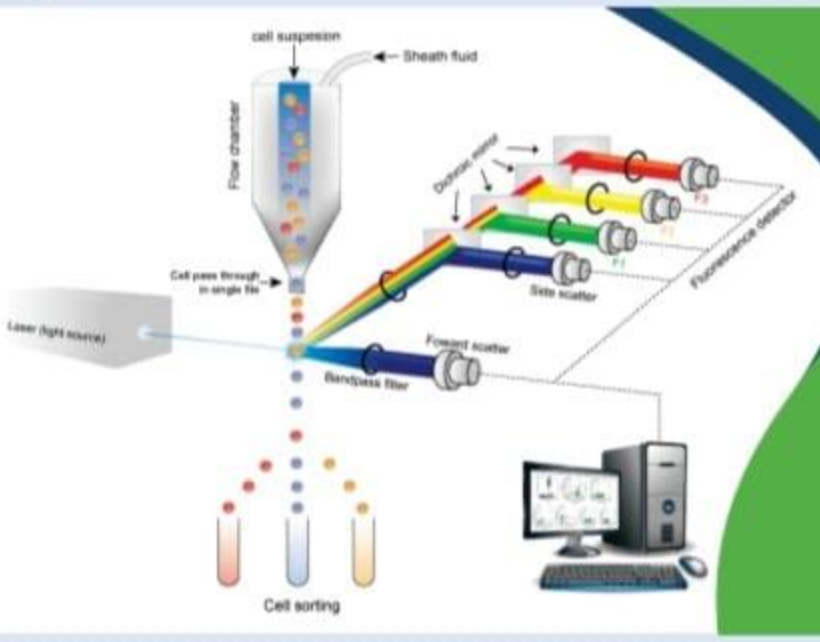
FLOW CYTOMETRY

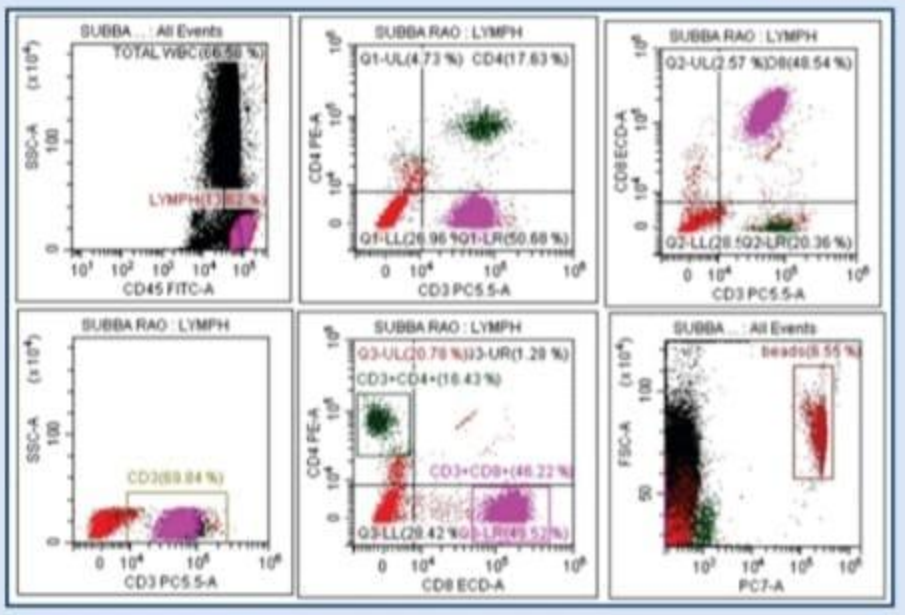
Sample Type: Peripheral Blood EDTA, Bone Marrow
Transit Temperature: 2-8 c

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- Flow cytometry is a technology that simultaneously measures and analyzes multiple physical characteristics of single cells, as they flow in a fluid stream through a beam.
- Hydrodynamics focusing produces a single stream of particles.
- Laser light source is used to excite fluorochromes at a particular wavelength & re-emit leading to fluorescence emission.
- Three laser Blue, Red and Violet and 13 channel fluorescent dyes used are FITC, PE, ECD, PC5.5, PC7, APC, APC-A700, APC-A750, PB450, KO525, violet610, violet660 & violet780.
- More than one fluorochrome can be used to analyze several parameters of sample at one time (MULTIPARAMETRIC ANALYSIS).
- Optical to electrical data coupling is done by the instrument using photomultiplier tubes (PMTs).
- Analysis of flowcytometric data is done using suitable gating strategy to visualize the cells of interest while eliminating unwanted particles i.e. dead cells or debris.
- Various combinations of fluorochrome labeled antibodies are used to make dot plots for analysis.
- For analysis, blood, bone marrow aspirate body fluids & tissue sample can be used.
- Sodium heparin is the preferred anticoagulant for longer stability (upto 72 hrs).
- The applications of Flowcytometry in diagnosis include the analysis of leukemia & lymphomas, detection of Minimal Residual Disease (MRD), stem cell enumeration, immunologic monitoring of HIV infected individuals, detection of autoantibodies, DNA content analysis, Immunodeficiency disease and assessment of structural and functional properties of erythrocytes, leucocytes and platelets.



ACUTE LEUKEMIA PANELS

1. **Acute Leukemia Comprehensive Diagnosis**

   The panel is designed to classify acute leukemia into ALL or AML with further subtyping.

   B cell markers: CD19, CD20, CD22, sIgM,

   T cell markers: CD3, CD4, CD5, CD7, CD8, CyCD3,

   Myeloid & Monocytic markers: CD13, CD14, CD16, CD33, CD36, CD64, CD11b, MPO,

   Others: CD10, CD34, TdT, HLA DR, CD45, CD117.

2. **AML Characterization Panel**

   The panel is designed for subtyping of confirmed AML cases.

   Myeloid & monocytic markers: CD13, CD14, CD16, CD33, CD36, CD64, CD11b, MPO,

   Others: CD117, CD34, HLA DR, CD45.

3. **ALL Characterization Panel**

   ALL panel is designed to distinguishing ALL from acute myelogenous leukemia (AML) and chronic lymphoproliferative, and for further subtyping of confirmed ALL cases

   B cell markers: CD19, CD20, sIgM,

   T cell markers: CD3, CD4, CD5, CD7, CD8, CyCD3,

   Others: CD10, CD34, CD38, CD45, TdT.

CHRONIC LYMPHOPROLIFERATIVE DISORDER (CLPD) PANELS

1. **CLL DIAGNOSTIC PANEL**

   The panel is designed for the confirmation and subtyping of CLPDS.

   T-cell Markers: CD2, CD3, CD4, CD5, CD7, CD8,

   B-cell Markers: CD19, CD20, CD23, CD79b, CD200, SIgM, Kappa, Lambda, FMC-7,

   Others: CD10, CD25, CD38, CD103, CD11c, CD45, HLADR, ZAP-70.

2. **LYMPHOMA PANEL**

   Lymphoma panel give confirmation and subtyping of CLPDs

   T- cell Markers: CD2,CD3,CD4,CD5,CD7,CD8.

   B-cell Markers: CD19, CD20, CD23, CD79b, CD200, SIgM, Kappa, Lambda, FMC-7.

   Others: CD10, CD25, CD103, CD11c, CD45, HLADR.

3. **HAIRY CELL LEUKEMIA PANEL**

   Hairy Cell Leukemia is a clonal B-cell malignancy caused due to the abnormal growth of B cells which develop hair like cytoplasmic projections on the surface and look "hairy" under a microscope.

   The pannel includes : CD19, CD20, Kappa, Lambda, CD25, CD103, CD11c, HLADR, CD45.

MULTIPLE MYELOMA PANEL

- Multiple myeloma (MM) is a plasma cell neoplasm characterized by proliferation of monoclonal plasma cells.
- MM panel is for confirmation of clonality and abnormal antigen expression in neoplastic plasma cells.
- MM pannel includes-CD19, CD20 CD38, CD45, CD56, CD138, Cy.Kappa, Cy.Lambda.

LEUKEMIA LYMPHOMA COMPREHENSIVE DIAGNOSIS PANEL

- Open panel for cases in which diagnosis of acute leukemia vs CLPDs is not clear on morphology or clinical findings.
- Stepwise approach is taken and comprehensive diagnosis is given.

PAROXYSMAL NOCTURNAL HEMOGLOBINURIA (PNH)

- Paroxysmal nocturnal hemoglobinuria (PNH) is a rare stem cell disorder characterized by the triad of intravascular hemolysis, peripheral blood cytopenias due to bone marrow failure and predisposition to venous thrombosis.
- The acquired defect in PNH is the loss of Glycosylphosphatidylinositol (GPI) anchored protein expression by hematopoietic stem cells. GPI linked Epitopes are easily detected on peripheral blood cells and can be accurately evaluated by Flowcytometry.

CD 34

- The hematopoietic stem cells (HSCs) in marrow and peripheral blood which are responsible for multilineage engraftment in the transplant setting express the cell surface marker CD34.
- Flowcytometry enumeration of CD34 cells provides a rapid means of measuring the clinically useful surrogate marker of graft adequacy in all sources of HSCs.

CD4/CD8 ENUMERATION

- The CD4/CD8 ratio is a reflection of immune system.
- The enumeration of CD4 and CD8 positive cells, surrogate markers for HIV disease progression, is helpful in management and follow up of immunocompromised HIV positive patients.

T -B -NK CELL PANEL

- T-cells are involved in combating intracellular infections, cancer cells, and foreign tissue, B-cells give rise to the humoral immune system and NK cells play a role in defense against viral infections and tumors.
- These lymphocyte subsets can be discerned by the antigenic properties of cell surface (membrane) markers.
- The panel is used to assess the immune status of patients.

HLA B-27

- The presence of HLA B-27 antigen by Flowcytometry is strongly associated with Ankylosing spondylitis, a chronic inflammatory disease of the axial musculoskeletal system and a few other rheumatic disorders (Reiter's syndrome, Acute anterior uveitis and Inflammatory Bowel disease).
- HLA-B2-7 testing is routinely used to screen for Ankylosing spondylitis, since 90% of patients with Ankylosing spondylitis have the HLA B2-7 surface antigen compared to only 8% of healthy individuals.